

Self-Supervised and Contrastive Representation Learning on STL-10

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Abstract—This report compares three ResNet-18 image encoders trained without pretrained weights on STL-10: end-to-end supervised learning with 500 labeled images, rotation prediction using the 100,000-image unlabeled split, and SimCLR-style contrastive learning using 20,000 unlabeled images. The supervised model and the two self-supervised encoders were evaluated on all 8,000 test images. The SimCLR representation achieved the highest linear-evaluation accuracy at 48.48%, followed by supervised learning at 44.15% and rotation learning at 41.49%. Nearest-neighbor retrieval gave a different ranking on the three selected queries: supervised features had the highest class-match rate (80%), while rotation and SimCLR each reached 66.67%. These results show that a model’s classification accuracy and its local retrieval behavior are related but not identical.

Keywords—STL-10, ResNet-18, self-supervised learning, rotation prediction, SimCLR, contrastive learning, linear evaluation, nearest neighbors.

I. INTRODUCTION

Image representations are feature vectors that summarize useful visual information in an image. A useful representation places images with related content near one another and makes class prediction possible with limited labeled data. This experiment compares three ways to learn such representations: learning directly from labels, learning to predict image rotations, and learning to match two augmented views of the same image.

The same ResNet-18 architecture, initialized from scratch, was used in all three approaches; each method trained its own encoder. For the self-supervised approaches, the encoder was frozen before training a new linear classifier on the same small labeled subset. This linear-evaluation setup tests how much class-relevant information is already present in the learned features.

II. DATA AND EXPERIMENTAL SETUP

STL-10 contains 5,000 labeled training images, 100,000 unlabeled images and 8,000 test images across ten classes. The labeled subset was sampled with seed 1598, with 50 images per class (500 images total). The same subset was used for supervised training and both linear evaluations. Images were processed at 96×96 pixels and normalized using the STL-10 channel means and standard deviations.

The notebook records the Step 0 personal parameters as $SID4 = SEED = 1598$, $SLICE = 598$, $HP_ID = 2$, $CLS_A = 8$ (ship) and $CLS_B = 5$ (dog). Because the assignment did not specify an HP_ID -to-hyperparameter mapping, HP_ID did not change the model settings. All runs used a Tesla T4 GPU with PyTorch 2.11.0+cu128 and torchvision 0.26.0+cu128.

Approach	Training data and objective	Training setup
Supervised	500 labeled images; 10-way cross-entropy	ResNet-18 encoder and classifier trained end-to-end for 15 epochs; AdamW, learning rate 0.001, weight decay 0.0001; cosine schedule; random crop and horizontal flip.
Rotation SSL	100,000 unlabeled images; predict 0° , 90° , 180° or 270°	ResNet-18 plus four-way rotation head for 15 epochs; batch size 256; AdamW, learning rate 0.0003, weight decay 0.0001; mixed precision.
SimCLR	20,000 unlabeled images; match two views of each image	ResNet-18 plus an MLP projection head ($512 \rightarrow 512 \rightarrow 128$) for 20 epochs; batch size 128; cosine similarity, temperature 0.2; AdamW, learning rate 0.0003, weight decay 0.0001; mixed precision.

For SimCLR, each view was independently created with four augmentations: random resized crop, random horizontal flip, color jitter and random grayscale. The contrastive objective treated the two views from the same source image as a positive pair and other images in the batch as negatives. After pretraining, the projection head was removed. For rotation SSL and SimCLR, the encoder was frozen and only a 512-to-10 linear classifier (5,130 trainable parameters) was trained for 20 epochs on the 500 labeled images. The supervised model trained its encoder and classifier together.

III. RESULTS

Table I summarizes the test results. Each classifier was evaluated on all 8,000 test images. The supervised model’s training accuracy reached 75.40%, compared with 44.15% test accuracy, indicating a substantial generalization gap. Rotation pretraining reached 64.08% accuracy on its pretext task, which is above the 25% chance level for four rotations. SimCLR’s contrastive loss decreased from 3.2136 to 1.5637 over 20 epochs.

Model	Test loss	Test accuracy	Neighbor match rate
Supervised	1.6781	44.15%	80.00% (12/15)
Rotation SSL + linear probe	1.6903	41.49%	66.67% (10/15)
SimCLR + linear probe	1.4308	48.48%	66.67% (10/15)

TABLE I. TEST CLASSIFICATION AND NEAREST-NEIGHBOR SUMMARY

IV. NEAREST-NEIGHBOR ANALYSIS

For each encoder, 512-dimensional embeddings were extracted for the test set and normalized. The same dog (index 7570), ship (index 6996) and car (index 5961) images were used as queries. Cosine similarity was computed between each query embedding and all test embeddings; the query itself was excluded before retrieving the five nearest images.

Query	Supervised matches	Rotation SSL matches	SimCLR matches
Dog	3/5: dog, deer, cat, dog, dog	0/5: deer, deer, horse, deer, horse	3/5: horse, dog, dog, horse, dog
Ship	5/5: all ship	5/5: all ship	5/5: all ship
Car	4/5: truck, car, car, car, car	5/5: all car	2/5: car, car, truck, truck, truck

Across these 15 retrieved images per model, supervised features achieved 12 same-class matches (80%). Rotation SSL and SimCLR each achieved 10 matches (66.67%). All three encoders retrieved five ships for the ship query. The largest differences appeared for dog and car: rotation features confused the dog query with deer and horse, while SimCLR’s car query included two trucks. These examples suggest that some nearby images shared broad visual characteristics or related object shapes even when their class labels differed.

The neighbor summary is based on only three queries, so it should be treated as a qualitative check rather than a comprehensive retrieval benchmark. Its ranking does not contradict the linear-evaluation accuracy ranking: classification accuracy aggregates predictions over all 8,000 images, while the neighbor match rate measures the local top-five neighborhood for three selected examples.

V. DISCUSSION

SimCLR performed best on test classification, reaching 48.48%. Its two-view objective encouraged the encoder to retain information shared across different crops, flips, color changes and grayscale versions. This likely produced features that transferred better to the linear classifier than the rotation objective did. The training loss decreased steadily, supporting that the contrastive objective was being optimized. However, the 66.67% match rate on the selected neighbor queries shows that this benefit did not make every local neighborhood class-pure.

The supervised model was second in test accuracy and first in the selected-query neighbor check. Direct class labels provide a task-aligned signal, but 500 examples are limited for training all 11.18 million parameters in ResNet-18. The 31.25 percentage-point difference between training and test accuracy is consistent with overfitting. More labeled examples would be the most direct improvement; stronger regularization, more varied augmentation, mixup or label smoothing, and careful early stopping could also reduce the gap.

Rotation SSL used five times as many unlabeled images as SimCLR, but its linear-evaluation accuracy was lowest at 41.49%. Predicting orientation can teach edges, shapes and spatial structure, but solving the rotation task does not always require learning features that separate object categories. The dog query retrieved no dogs, although ship and car retrieval were strong. Longer pretraining and a richer pretext objective, such as combining rotation prediction with contrastive learning, could improve class-relevant features. Fine-tuning some encoder layers after linear evaluation could also help, though it would no longer be a strictly frozen linear probe.

SimCLR could be improved by training longer, using more of the 100,000-image unlabeled split, increasing the batch size if memory permits, and tuning augmentation strength and learning-rate scheduling. Larger batches provide more negative examples for the contrastive objective. These changes may improve representation quality, but they would require more compute and should be measured rather than assumed to help.

VI. LIMITATIONS

The comparison uses one seeded labeled subset, one training run per method, and a limited set of training durations. Results may vary with a different subset or random initialization. The nearest-neighbor analysis uses only three queries, so it is illustrative and does not establish overall retrieval performance. The training and test sets were kept separate for supervised fitting and evaluation; the test labels were used only to describe retrieved neighbors after the embeddings had been produced.

VII. CONCLUSION

On this run, SimCLR produced the strongest classification representation, with 48.48% test accuracy, followed by supervised learning at 44.15% and rotation SSL at 41.49%. The nearest-neighbor check offered a complementary view: supervised features matched the query class most often overall, while all three models retrieved the ship query consistently. The results demonstrate that the best model depends on the evaluation: SimCLR led on full-test linear classification, while supervised features gave the most class-consistent neighborhoods among the three chosen queries. More labels, improved regularization, longer or broader self-supervised pretraining, and a larger retrieval evaluation are practical next steps.

APPENDIX A SELECTED NEAREST-NEIGHBOR VISUALIZATIONS

Each row shows one query and its five highest-cosine-similarity neighbors in the test embedding space. Titles above the retrieved images give the class label and cosine similarity. The same dog, ship and car queries were used for every encoder.

Fig. A1. Supervised encoder: query images and top-five nearest neighbors.

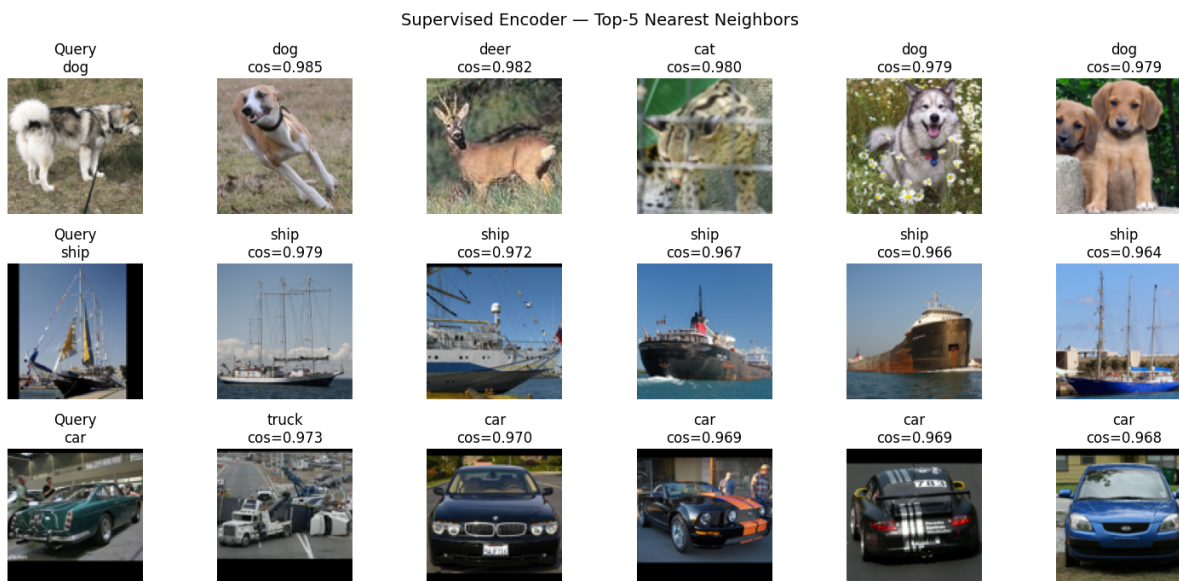


Fig. A2. Rotation SSL encoder: query images and top-five nearest neighbors.

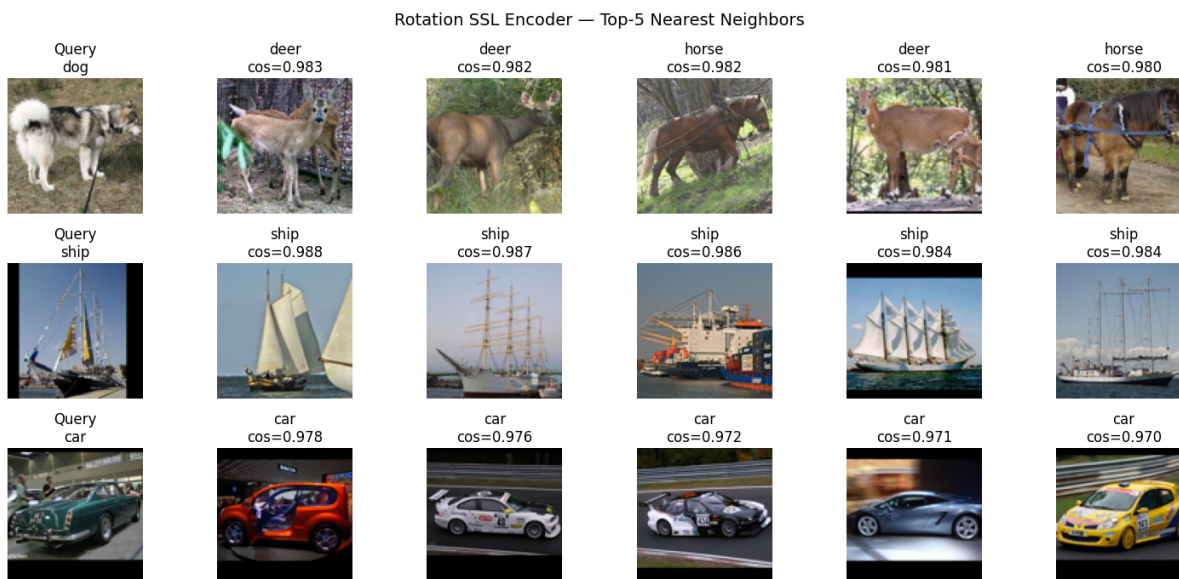


Fig. A3. SimCLR encoder: query images and top-five nearest neighbors.

